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Chapter 7

Style Production Overview



Introduction to Style Production

Style Production has been developed to meet specifically the production needs of organisations within the clothing and apparel, footwear and soft furnishings industries.

Although the products that these organisations produce are diverse, the manufacturing requirements are similar. They require:

- A horizontal sequence of operations defined in the route:



- A means of maintaining information about products at a generic level.
- Two levels of items, although it is possible to have more than two levels:
 - Styles. The finished goods.
 - Materials. The materials that go into producing the styles.

Style Production provides a means of maintaining common information for the multiple variations of a style in one action. That is, you do not have to enter information about each style variant separately.

In addition, Style Production offers the option to produce some style variants on different machines and/or using different materials.

In common with other JBA applications, you can operate a single copy of Style Production for any number of companies.

Using control parameters held on the Company Profile and the Organisational Model, you can configure the characteristics of each company.

Brief Overview of Style Production

Style Production comprises four integrated applications, as shown below:

Application Code	Application	Application Product Guide
P1	Style Production Definition Management	Style Production Definition Management
P2	Master Production Scheduling	Style Production Planning
P3	Material Requirements Planning	Style Production Planning
P4	Style Production Control	Style Production Control

The Company Profile governs these core applications.

Style Production Definition Management (PDM)

Style Production Definition Management (PDM) offers a self-contained environment for defining and managing products, processes, resources and costs.

Use the maintenance functions of PDM to set up information that is critical to the way in which Style Production functions. Once you have set up this data, PDM can generate product costs (both standard and non-standard) and provide an extended analysis of product definitions and processes.

You can also choose to run PDM stand-alone.

Forecasting

Forecasting belongs to the PDM application.

Forecasting allows you to forecast demand down to the level of individual variants. For example, for clothing you can forecast down to the level of style/colour/size.

You have the option to group products into product families.

You can input the forecast demand into MPS and MRP processing.

Master Production Scheduling

Master Production Scheduling (MPS) provides a top level master production schedule for manufacturing styles.

Material Requirements Planning

Material Requirements Planning (MRP) provides a detailed material plan for your business.

Production Control

Production Control controls and maintains all aspects of the production cycle.

The cycle begins with the release of suggested production orders from MPS. Alternatively, you can create production orders yourself. The cycle continues through material issuing and work-in-progress bookings. At the end, Inventory receives the finished goods.

Production Control also provides comprehensive enquiries and reports to monitor the progress of individual production orders and the performance of factory/production units.

Interfaces

There is one mandatory interface to Style Production and that is Style Inventory Management.

The following interfaced applications are optional:

- Style Purchase Management. However, if you intend define subcontracted operations on route/BOMs, then Style Purchase Management is mandatory.
- Style Sales Order Processing.
- Warehousing.

To use certain facilities available within the system, additional optional applications must be 'live'. This is especially true within Style Production Planning which looks to other areas of a fully integrated production system for external demand and supply information.



For a full description of each interface, please refer to the application's own product guide.

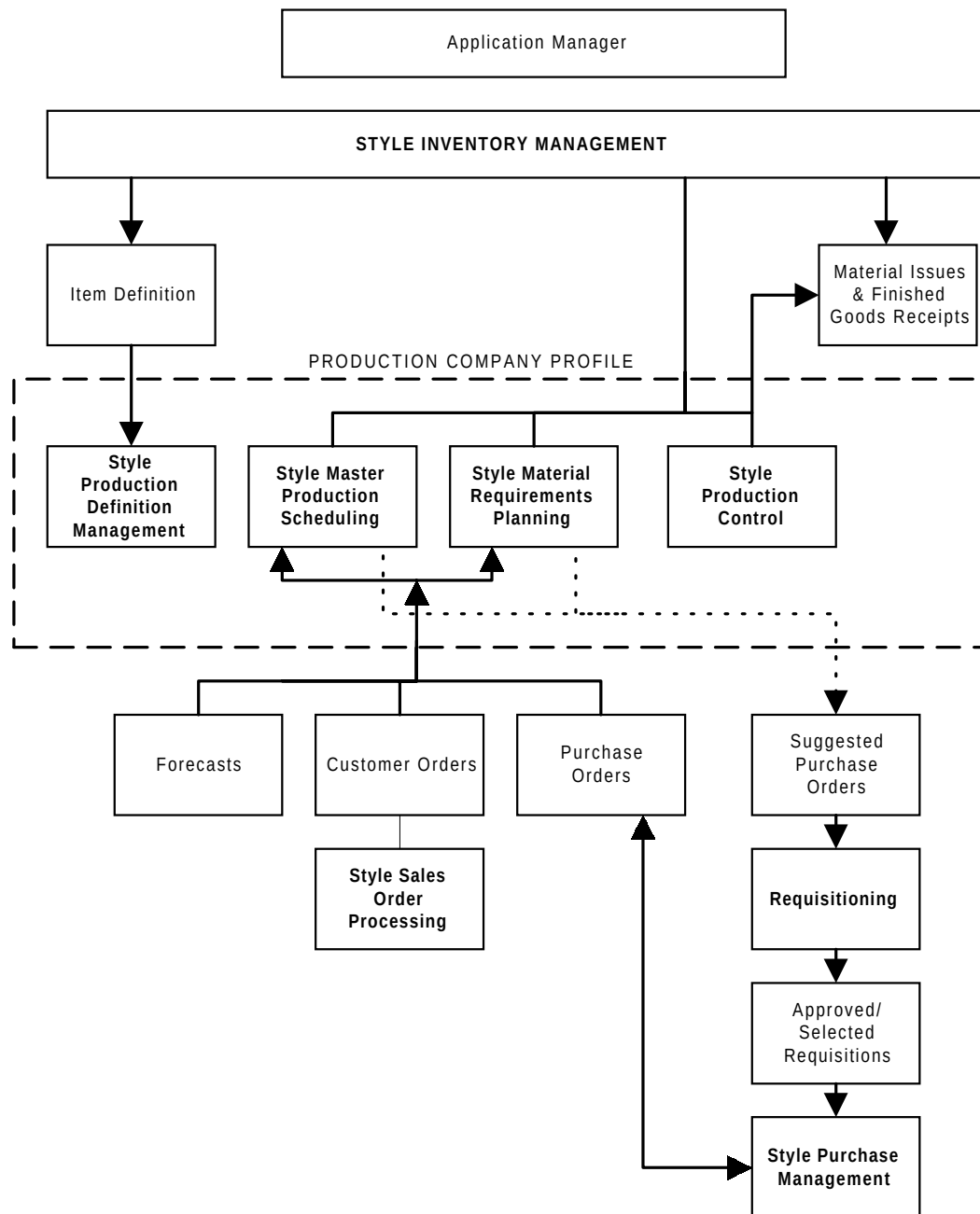


Figure: The Style Production System

Optional Interfaces

Style Production can function fully when interfaced to Style Inventory Management alone. More commonly, Style Production operates within a larger, integrated systems environment.

Integration with the following applications provides additional resources for management of a sophisticated, closed loop production environment.

Style Purchase Management

Style Purchase Management and its associated Requisitions function provides a two way interface to Style Production:

- Style Purchase Management provides MPS/MRP with details of current material supply information.
- MPS/MRP generates suggested purchase orders, which when confirmed, raise purchase requisitions, which in turn are converted to firmed purchase orders.

You can arrange to create purchase orders automatically for subcontracted operations.

Complementary to running Style Purchase Management is the interface to Accounts Payable (Purchase Ledger), where you must create all supplier records.

Subcontract operations depend on Purchase Management, as each operation designated subcontracted must be assigned a supplier, which in turn must be valid within Purchase Management and associated with a live account in Accounts Payable.

Style Sales Order Processing

Style Sales Order Processing provides a two way interface to Style Production:

- Style Sales Order Processing provides MPS/MRP with details of current sales demand information.
- Style Production provides Sales Order Processing, via Style Inventory Management, with current details of how supply is intended to meet sales demand.

Style Production depends on Style Sales Order Processing for the supply of demand information only.

To create a forecast of sales demand, use Style Production. Alternatively, use the System 21 Forecasting application to extract information about styles from Style Inventory and sales history from Style Sales Analysis.

Warehousing

Warehousing is an extension of Style Inventory Management stockrooms. You can use Warehousing to designate location details for materials and styles within a stockroom.

You can use Warehousing to:

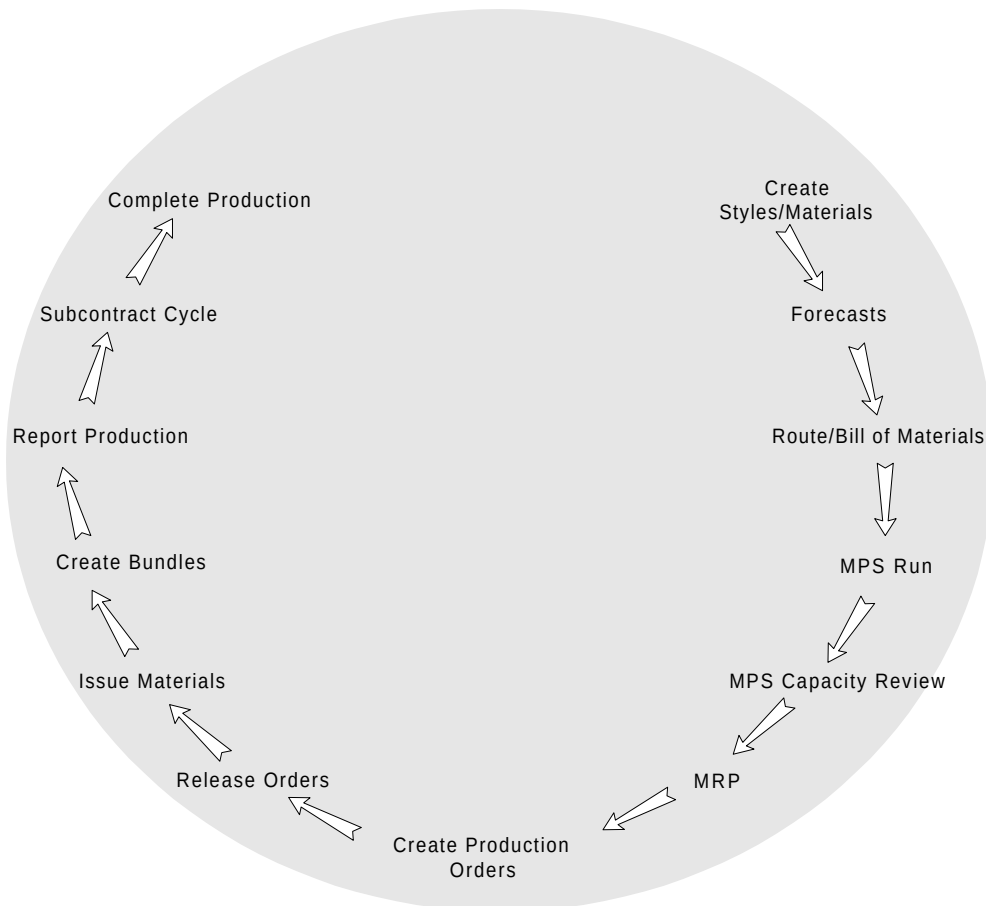
- Satisfy material requirements generated by production orders.
- Provide a receiving location for finished goods.

Warehousing is not a pre-requisite of Style Production.

The Business Cycle

Introduction

The following diagram gives an overview of the business cycle:



Create Styles And Materials

Styles are the end products that your business sells to customers. Materials are the materials, such as fabric, trims and packaging, that make up the styles.

With Style Production, you can enter information about a style or material in one go. That is, you do not have to enter information for each colour and size variant separately.

Style views styles and materials as multi-dimensional. The first dimension is always the style or material itself. You can use up to four dimensions per style, as follows:

- Style.
- or
- Style / Colour.
- or
- Style / Colour / Size.

or

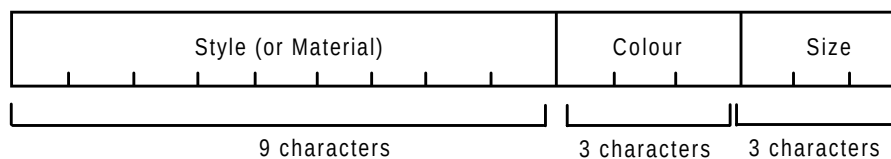
- Style / Colour / Size / Fit.

Dimension	Characteristic	Such as
Style (or Material)	-	Skirt
2nd Dimension	Colour	Navy, Black, Green
3rd Dimension	Size	10, 12, 14
4th Dimension	Fit	Short, Medium, Long

Therefore, each style and material can have colour variants and up to two types of size variant. Size variants can take many forms such as:

- Trousers: inside leg and waist size.
- Curtains: height and drop.
- Shoes: size and fit.

Each style and material has a unique identifier, known as the style code or material code. Style variants extend this identifier to include codes for the colour and size(s). A variant code has the following format:



Examples of typical codes for style and material variants are shown below:

	Dimensions	Variant	Typical style/material code
1 dimensional:	Style	There are no variants: the label has no colour or size characteristics	LABEL005
2 dimensional:	Style/Colour	Green scarf	SCARF01 GN
3 dimensional:	Style/Colour/Size	Large black tee shirt	TS15G BLK L
		Red fabric 140cm	FAB001 RD 140
4 dimensional:	Style/Colour/Size/Fit	Navy skirt size 12 short	SK010 NV 12S

Forecasts

You can enter forecasts at the following levels:

- Style.
- Style / Colour.
- Style / Colour / Size.
- Style / Colour / Size / Fit.
- Product Family.

After entering a forecast, you can spread the forecast as shown below:

FROM	TO
Product Family:	Style
	Style / Colour
	Style / Colour / Size
Style:	Style / Colour
	Style / Colour / Size

Routes / Bills Of Material

A production route can produce multiple variations of styles such as different colours and sizes of jumpers, shirts, pairs of shoes. A production route can also produce a component for use in the manufacture of a finished style.

A production route defines:

- The operations required to make a style.
- The materials necessary to complete the operation.
- The machines at which the work is carried out.

Each style can have one or more routes. On each route, you can:

- Match style colours to material colours.
- Change material quantities per style size.
- Change operation details for different style colours and sizes.

Style Production caters for multiple plants. If you are making a particular style in more than one plant, multi-plant planning can tell Style Production to shift the load from one factory to another factory. This facility is useful where a factory already has a full load.

You can experiment with model planning routes that do not affect live production. You can create model planning routes from new, or you can copy a live production route to a model planning route. When you are happy with the changes you have made to a model planning route, you can copy the route back to replace a live production route.

You can control access to live production route maintenance. For example, you can authorise a user to maintain model planning routes with no access to live production routes.

MPS Run

You can run MPS centrally or multi-plant or from a single plant.

MPS always performs a full regenerative run.

An MPS run carries out the following activities:

1. Examines sales orders, purchase orders, production orders, stocks and forecasts.
2. Nets off sales orders against forecasts to produce a net demand.
3. Works out when styles are required in terms of time buckets.
4. Suggests new/changed production orders and purchase orders. New orders can be based on sourcing rules and planning overrides.
5. Through capacity planning, determines whether there is sufficient capacity to action the suggested production orders.
6. Generates machine and labour loading.



Tip: You can arrange for MPS to consider purchased styles also, without running MRP. MPS can suggest purchased orders which you then need to activate via Requisitioning in Style Purchase Management.

MPS Capacity Review

After an MPS run, you can review MPS and you can review capacity loading.

During the MPS review, you can:

- Review production planning at style level.
- From a suggested production order, drill down to individual style variants.
- Change the route, date and quantity on a suggested production order.
- Change the status of a suggested production order to planned or confirmed.
- Review the impact of your changes on capacity.

In the garment industry, the most critical planning element is labour skills. Therefore, Style Production differs from Standard Production by including skill-based planning.

During a capacity review, you can:

- Review the labour and machine loads that MPS generates.
- If you see an overload, move a suggested production order before you firm up the order.
- Consider how the move affects labour and machine load.
- Manipulate the data again.

- When you are happy, firm up the order and save the model as a different model.

MRP

You can run MRP either before or after firming up MPS orders.

You can perform a full regenerative run or a selective run. You can also MRP to consider only the changes that have occurred since the last MRP run.

Therefore:

1. Run MPS.
2. Stop and manipulate the data.
3. Run MRP.



Note: MRP does not carry out capacity planning.

Create Production Orders

Create production orders manually or process production orders suggested by MPS.

MPS suggests production orders for styles broken down to style variant level. MPS calculates supply at style variant level, but presents information at style summary level so that you can manipulate the information at style level.

Release Orders

To make sure you have all the materials you require, you can carry out a trial kit. Within Style Production, you can make lot control automatic. You can also arrange to create bundles at order release.

If a subcontractor is responsible for any of the operations, then Style issues a purchase order.

Issue Materials

When you issue materials, you can:

- Accept or change the lots that Style allocates.
- Issue substitute materials.
- Record unplanned issues.
- Transfer materials to a subcontractor, if you are outworking the cutting operation.

Create Bundles

A bundle is a large plastic bag containing all the fabrics and other materials (eg buttons) that go into producing a garment.

You can create bundles in two ways:

- Automatic. You can set up a number of different bundle types for which you specify the quantity to go in each bundle. You can allocate a bundle type to a specific route and arrange for Style to generate bundles automatically.
- Manual. You can decide on the bundle size when you create the bundle.

The bundling process can print barcoded bundle tickets. Style Production uses bundle tracking to track the garment from one operation to another.

Report Production

There are various ways of entering time spent in production:

- Operator bookings. Actual time is keyed in.
- Timesheet bookings. Actual time is calculated.
- Scan in barcodes on bundle tickets. Alternatively, key in the bundle number. Actual time is calculated.

Subcontract Reporting Cycle

Where several subcontractors are carrying out consecutive operations, you can define the operations separately. Style can report on progress from operation to operation.

When you release the production order, Style looks down the list of subcontracted operations and generates a subcontractor purchase order.

When you report the operation previous to the subcontracted operation, Style creates a subcontract shipper against which you can ship out the goods. If several subcontractors are on the route, you can issue either one or several subcontract shippers.

When you receive the goods back, Style generates a good received note for the purchase order. You can then carry on booking against the next operation on the route.

Complete Production

You can complete production when you book in the final operation. If you book in the full quantity, Style automatically completes the production order.

Even if the quantity you book in is short, you can close the order.

You can re-open a production order.

At the end of the production cycle, the completed products pass into Style Inventory.

The Critical Path to Production

This section provides an overview of the critical path to follow in setting up a Style Production company, managing data and ultimately shipping your finished styles to a customer.

The following path is relevant to a fully comprehensive integrated system that is running all of the Style Production applications and also the following applications:

- Style Inventory Management.
- Style Sales Order Processing.
- Style Purchase Management and, by implication, Accounts Payable and Purchase Ledger.

If your system is not as complex as this, then you may skip non-relevant stages and data definitions.

Each of the following stages is numbered consecutively in the order in which data definition should take place.



Tip: Use the critical path listed below as a guide only. You may be able to jump to another task before all previous tasks are complete.



Tip: You may want to hold discussions with your system consultant with regard to an implementation procedure specific to your requirements.



Tip: For a full description of each data element in the critical path, refer to the associated product guides that accompany each System 21 application.

The Critical Path to Production			
	Description	Applica- tion	Menu Option
	<u>Prior To Style Production</u>		
1	Create a new company(s) via the Applications Manager	Admin Functions	
2	Authorise user(s) to new company in Style Distribution	Admin Functions	
3	Copy company in Style Distribution	STIN	Copy Company
4	Maintain Style Sales Order Processing, Company Profiles. Set the Style Production in Use flag to 1 (yes)	STIN	Sales Order Processing Company Profile

5	Maintain Style Inventory Management, Company Profiles. Create stockroom records, including work-in-progress (WIP) locations and subcontractor materials stockroom	STIN	Inventory Management company profile
6	Maintain Style Purchase Management Company Profiles	STIN	Purchase Management Company Profile
7	Create suppliers, including subcontractors	PL (Accounts Payable)	Names and Addresses
8	Create colours for each company	STIN	Colours
9	Create Inventory Management calendars	STIN	Calendars
10	Create Inventory Management periods	STIN	Period/Week End Control
11	Create search families	STIN	Search Family Codes
12	Create sizes	STIN	Sizes
13	Create size masks	STIN	Size Masks
14	Create inventory descriptions	STIN	Descriptions
	Style Production		
15	Authorise user(s) to Style Production	Admin Functions	
16	Maintain Style Production Company Profiles	P1	Maintain Company Profile
17	Create Organisational Models	P1	Organisational Models
18	Create Company parameters	P1	Maintain Parameter File
19	Create styles and materials	P1	Style/Material Details
20	Create cost centres	P1	Cost Centres
21	Create calendars	P1	Calendars
22	Create shift profiles	P1	Shift Profiles
23	Create departments	P1	Departments
24	Create machines	P1	Machines
25	Create work centres	P1	Work Centres
26	Create labour skills	P1	Labour Skills
27	Create labour profiles	P1	Labour Profiles
28	Create bundle ticket types	P1	Bundle Ticket Types
29	Create resource capacity profiles	P1	Resource Capacity Profiles

30	Create critical resources	P1	Critical Resources
31	Create standard operations	P1	Standard Operations
32	Create style route/BOMs	P1	Style Route/Bill Of Material
33	Create price list profiles	STOE	Price/discount profiles
34	Create customers	STOE	Customers
35	Create price lists	STOE	Price Lists
36	Create discount lists	STOE	Discount Lists
37	Create depot profiles	STOE	Depot Profiles
38	Enter sales orders	STOE	Order Entry
39	Create MPS planning models, part one	P2	Maintain Model Stockrooms
40	Create MPS planning models, part two	P2	Maintain Reporting Profile
41	Create style forecasts	P1	Maintain Style Forecast
42	Spread forecasts to variants	P1	Generate Variant Forecast
43	Run MPS	P2	Run Production Scheduling
44	Review MPS	P2	Review Production Schedule
45	Review capacity	P2	Capacity Review
46	Confirm suggested orders	P2	Confirm Suggested Orders
47	Create MRP planning models, part one	P3	Maintain Model Stockrooms
48	Create MRP planning models, part two	P3	Maintain Reporting Profile
49	Run MRP	P3	Run Material Planning
50	Review MRP	P3	Review
51	Create purchase orders from suggestions, part one	STPM (Requisitioning)	Create Requisitions from MPS/MRP
52	Create purchase orders from suggestions, part two	STPM (Requisitioning)	Requisitions Confirmation
53	Create purchase orders from suggestions, part three	STPM (Requisitioning)	Requisitions Authorisation
54	Create purchase orders from suggestions, part four	STPM (Requisitioning)	Create Purchase Orders
55	Create purchase orders from suggestions, part five	STPM	New Orders

56	Enter purchase orders manually, part one	STPM	Order Entry
57	Enter purchase orders manually, part two	STPM	New Orders
58	Enter raw material receipts	STPM	Goods Receiving
59	Transfer purchasing costs	P1	Purchased Items Re-Cost
60	Full cost roll up	P1	Item Master File Re-Cost
61	Create activity types	P1	Control File Maintenance
62	Create operators	P1	Operators
63	Create teams	P1	Teams
65	Define user authority levels	STPM	User Authority
66	Create manual production orders	P4	Maintain Orders
67	Allocate lot materials	P4	Order Progress
68	Release production orders	P4	Release Orders
69	Issue materials	P4	Issue Materials
70	Report (book) production	P4	Production Order Booking or Operator Booking or Manual Timesheet Booking
71	Enter subcontractor WIP shippers	P4	Subcontractor WIP Shipper
72	Progress subcontractor work	P4	Progress At Subcontractor
73	Receive from subcontractor	P4	Receive from Subcontractor
74	Receive into finished goods	P4	Production Order Booking or Operator Booking or Manual Timesheet Booking
75	Enter supplier invoices, part one	PL (Accounts Payable)	Item Entry
76	Enter supplier invoices, part two	PL (Accounts Payable)	Confirm Log Entries
77	Enter supplier invoices, part three	PL (Accounts Payable)	Volume Log to Ledger Transfer

78	Match invoices to GRNs	STPM	Match Invoices to Receipts
79	Despatch to customer	STOE	Confirmation of Despatch